Titel: **The Padma Corridor as a Spatial Transformer: A Remote Sensing Analysis of LULC Change of Southern Bangladesh**

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1.Introduction

The southwestern part of Bangladesh has historically been physically and economically detached from the rest of the country due to the geographical barrier posed by the Padma River. However, the opening of the Padma Multipurpose Bridge in 2022 created a fast and reliable road connection that is now reshaping both the transportation and land-use patterns of the region. Faster and more dependable access to Dhaka and the Mongla Port is making Southwestern districts more attractive to investors. This development is already underway; two new Export Processing Zones (EPZs) have been established, and Ready-Made Garment (RMG) products now have a streamlined route through Mongla Port (Business Standard, 2023).

To accurately measure the true impact of the Padma Corridor, it is necessary to assess both the physical transformation of the land cover and the functional accessibility gains, viewed as the potential for increased interaction. A Land Use/Land Cover (LULC) change map will help identify which regions are experiencing higher economic growth in terms of vegetation transitioning into impervious surfaces. Furthermore, an accurate and up-to-date map will facilitate sustainable land planning and a deeper understanding of human-environment relationships.

In the age of Artificial Intelligence (AI), Google Earth Engine (GEE) is highly effective for LULC classification because it provides free access to massive satellite image collections, such as Sentinel and Landsat imagery. GEE allows users to process these large datasets directly on Google's cloud servers, removing the barriers of requiring high-performance devices or downloading and storing vast amounts of data.

Selecting the appropriate satellite imagery depends on the objective, scale, and required accuracy of the research. For large-scale vegetation monitoring and climate studies, MODIS imagery is widely used because of its high temporal resolution (Zhang et al., n.d.). Thermal sensors from satellites such as Landsat are useful for studying land surface temperature and urban heat islands (Tran et al., 2017a). For LULC classification specifically, Landsat 8 and Sentinel-2 are

among the most utilized datasets (Praticò et al., 2021). Both offer free, long-term multispectral imagery; Landsat 8 provides a 30m spatial resolution with an extensive historical archive; ideal for long-term change detection—while Sentinel-2 provides a 10m resolution for higher spatial accuracy.

In LULC classification, the number and types of classes primarily depend on the research objectives, study area characteristics, and data availability (Tran et al., 2017b; Weng et al., 2004). However, vegetation, waterbodies, and impervious surfaces remain the fundamental classes, as they represent the major components of the Earth's surface essential for environmental and urban analysis. Similarly, there is no single "best" classification algorithm; effectiveness depends on image resolution, landscape complexity, and scale. Consequently, researchers commonly test different algorithms to select the one that yields the highest classification accuracy for their specific study area (Praticò et al., 2021).

Research Question

This study is an attempt to see the relationship between accessibility gain and LULC change.

1. Using LULC change and night light image analysis, identifying regions that gain more economic growth for 58 upazila of southwest Bangladesh.

2. Methodology

This study utilizes GEE for satellite image processing and analysis. Google Earth Engin is highly efficient for LULC change detection as it supports advanced machine learning algorithms such as Random Forest (RF), Support Vector Machines (SVM) for classification. Another advantage of GEE is it provides free access to massive satellite image collections and can handle multi-temporal and seasonal image analysis(Praticò et al., 2021). GEE platform combines image processing, classification, visualization, and export functions in one environment. GEE is the faster, easier, and more accessible for researchers and planners working on LULC mapping and environmental monitoring.

2.1. Satellite Data Selection

To accurately capture the spatial transformation driven by the Padma Bridge, this study prioritized high-resolution multispectral imagery. Initially, Landsat-8 imagery was considered to establish a 2015 baseline. However, its 30-meter spatial resolution was too coarse to differentiate the highly fragmented and complex landscape of the southern Bangladesh study area. Therefore, the study utilized Sentinel-2 imagery, shifting the baseline year to 2017. During preprocessing, image collections were carefully filtered to maintain the same time of year, taking seasonal variability into consideration.

2.2. Classifications

To measure the transition from land cover to land use, the study area was classified into three fundamental classes: Waterbody, Vegetation, and Impervious Surface. To ensure the most accurate results, training samples were collected independently for each year, 2017 and 2025.

2.3. Machine Learning Classification and Validation

The study evaluates two supervised machine learning algorithms: Random Forest (RF) and Support Vector Machine (SVM). To ensure accuracy, the training samples were divided into two subsets: 80% for training the model and 20% for validating it. Both the RF and SVM models were trained using the same training dataset and showed nearly similar performance. The validation error matrix revealed an overall accuracy of approximately 90.9% and a Kappa coefficient of 0.86. Ultimately, the Random Forest classifier was selected for the final analysis because it requires fewer training samples and is highly efficient for large-scale composite imagery.

2.4. Change Detection

After classifying LULC for 2017 and 2025 using the RF classifier, the raster files were further analyzed in ArcGIS. The Raster Calculator tool was used to identify categorical changes,

primarily the conversion of vegetation into impervious surfaces. To aggregate these changes within administrative boundaries, the Zonal Statistics tool in ArcGIS was used to calculate the changes for each upazila within the study area.

2.5 Change Index

A standardized measure of vegetation loss is needed to compare land use changes across different zones. Therefore, a standardized transition rate (vegetation-to-impervious conversion rate) was calculated by dividing the area of vegetation converted into impervious surfaces by the total original vegetation area. The total vegetation area included vegetation that remained unchanged, vegetation converted into waterbodies, and vegetation converted into impervious surfaces.

$$\bullet \ \frac{\text{Vegetation} \rightarrow \textit{Impervious}}{\textit{Vegetation (No Change)} + \textit{Vegetation} \rightarrow \textit{Water} + \textit{Vegetation} \rightarrow \textit{Impervious}}$$

2.6 VIIRS Nighttime Light Intensity Analysis

To validate LULC change as an indicator of economic growth, this study utilizes nighttime light (NTL) intensity data from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB). Nighttime light intensity has become widely used in economic geography and regional studies as a proxy for economic growth and economic activity, especially in areas where reliable economic data are limited or unavailable. Satellite-observed night lights can effectively capture economic growth because increased economic activity generally leads to higher consumption of electricity and artificial lighting. Changes in light intensity strongly correlate with changes in GDP growth (Henderson et al., 2009). The VIIRS image preprocessing was completed using GEE, and the exported raster files were further processed in ArcGIS. Using the Zonal Statistics tool, mean NTL values were extracted for both 2015 and 2025 across all 58 regions.

2.7 Accessibility Score

For each of the 58 upazilas, potential accessibility was calculated for 2025, and the change rate was measured using the Hansen gravity-based model. The results were later compared with the LULC change and NTL analyses.

3 Results and Discussion

3.1 LULC change (2017-2025)

Between 2017 and 2025, vegetation decreased by 109.41 sq. km in total, whereas impervious surfaces increased by 774.88 sq. km (Table 1). The results show a sharp decline in vegetation and a significant increase in impervious surfaces, indicating that human activity increased substantially by 2025.

Variable	2017 Area	2025 Area	Change
Waterbody	5792.3682	6076.9683	284.6
Vegetation	9043.1522	8933.745	-109.41
Impervious	2306.2413	3081.1237	774.88

Table-1: LULC Area Change Between 2017 and 2025

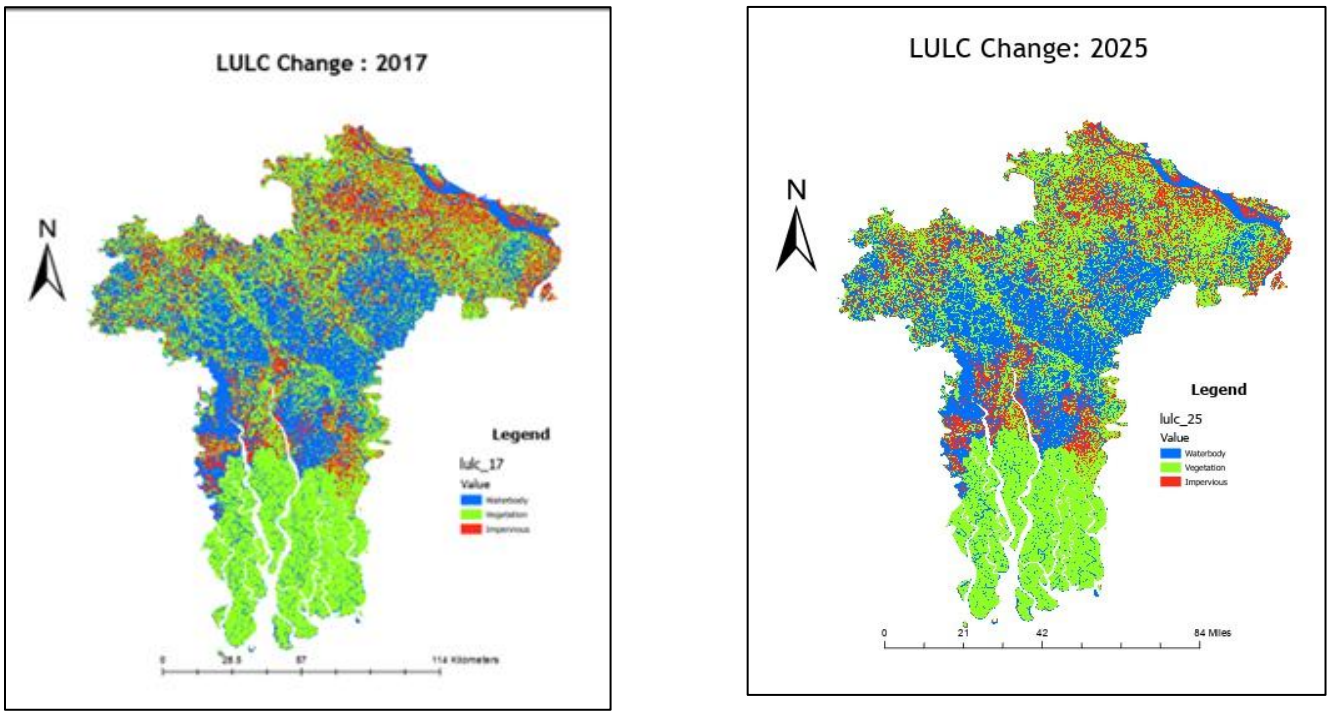


Figure-1: LULC Area Change 2017 and 2025

3.2 Rate

The rate analysis shows Land Use Land Cover (LULC) transformations across 58 upazilas in southern Bangladesh, focusing on the conversion rate from vegetation to impervious surfaces.

The analysis clearly indicates that upazilas within the Khulna Division show the highest levels of urbanization and economic growth. Batiaghata Upazila has the highest conversion rate at 31.17% (0.311). Other closely following upazilas are Paikgachha (22.48%) and Morrelganj (16.82%). Areas within the Khulna City Corporation (KCC), such as Kotwali (15.18%) and Sonadanga (10.54%), are also rapidly losing vegetation to newly developed impervious surfaces. Conversely, upazilas in the Dhaka Division, such as Sadarpur in Faridpur (8.51%) and Gosairhat in Shariatpur (6.99%), show relatively moderate to lower rates of LULC change. Several coastal and peripheral upazilas, such as Mongla (0.13%) and Koyra (0.48%), still maintain notably low conversion rates.

Rank	Upazila	District	Change Percentage (%)
1	Batiagahata	Khulna	31.17%
2	Paikgachha	Khulna	22.48%
3	Morrelganj	Bagerhat	16.82%
4	KCC (Kotwali)	Khulna	15.18%
5	Dumuria	Khulna	14.05%

Table-2: Ranking of Upazilas Based on LULC Change (2017–2025)

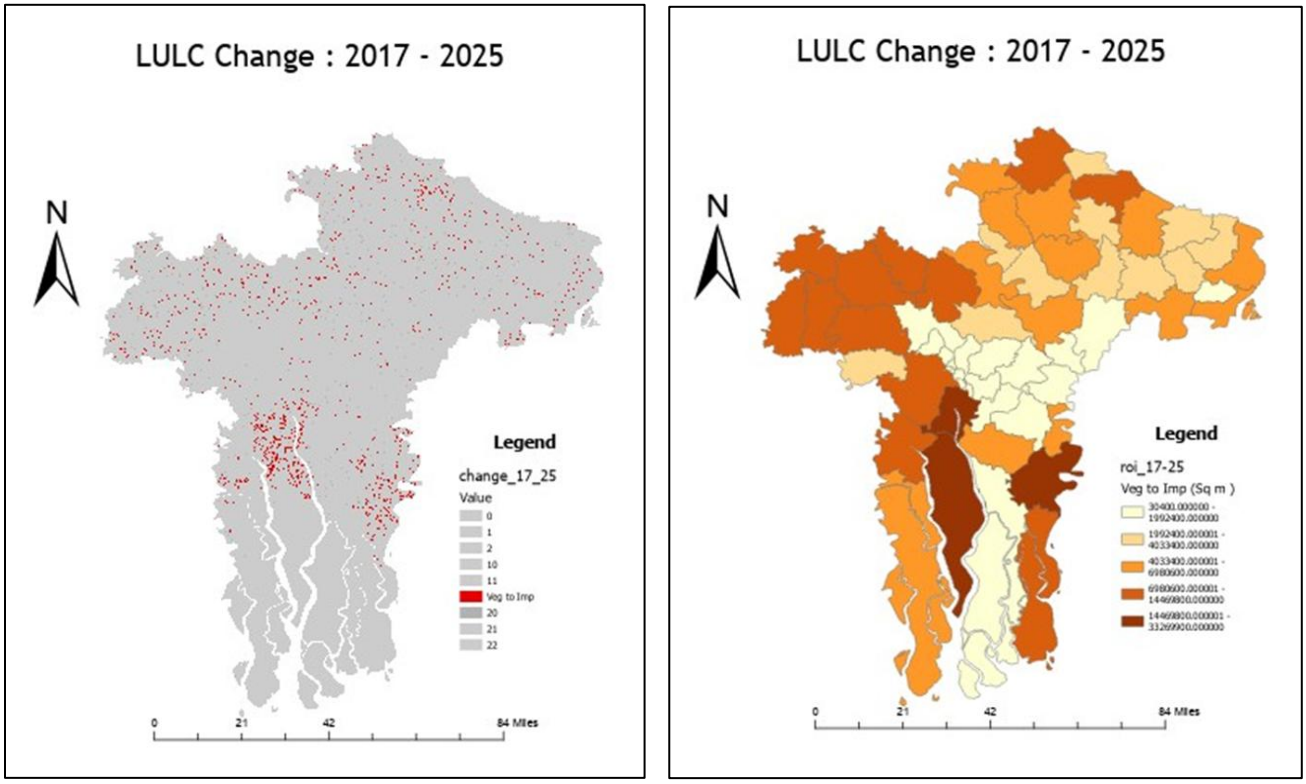
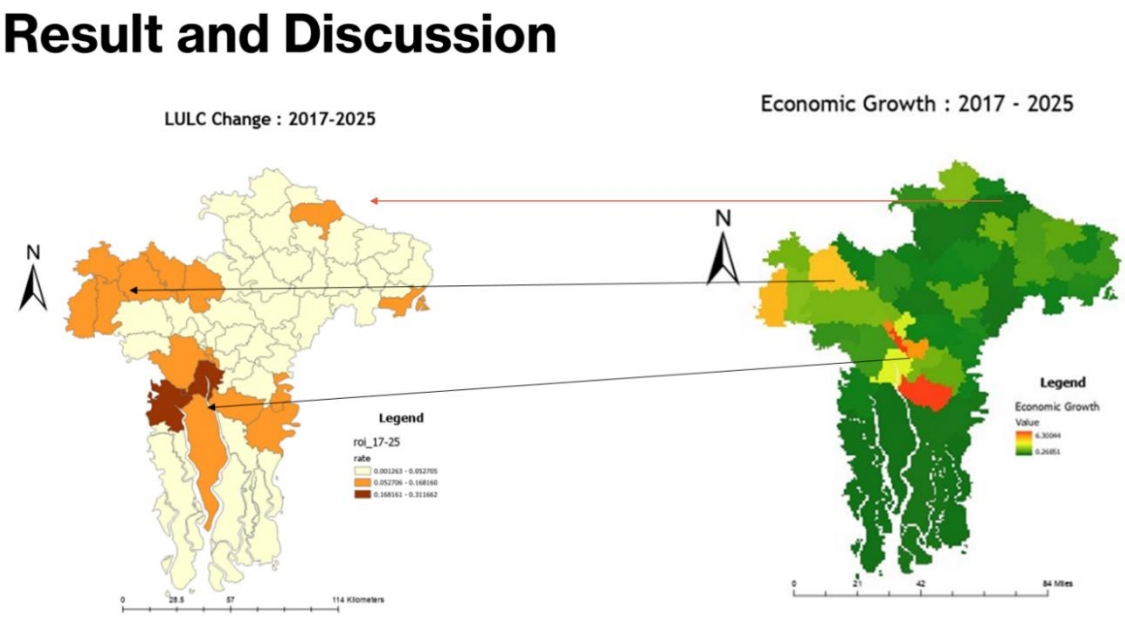


Figure-2: Changes in LULC between 2017 - 2025

Compare with Nightlight

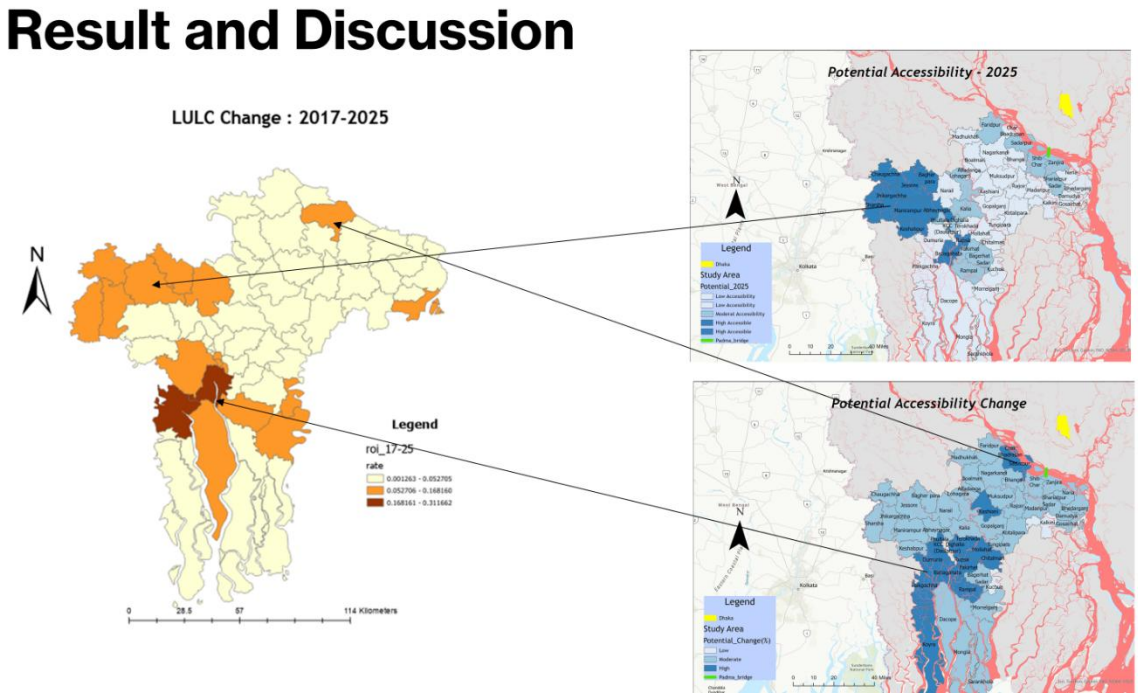
The results show a strong relationship between land cover change and economic growth between 2017 and 2025. Areas that experienced high conversion from vegetation to impervious surfaces also showed an increase in mean nighttime light intensity. This indicates that urban expansion and infrastructure development are closely linked with economic activities in the region. However, one major exception is the region highlighted on the map. Although this area experienced substantial vegetation-to-impervious surface conversion, it did not show a similar increase in nighttime light intensity. This is likely because large-scale construction activities are still ongoing in the region. Since many projects are not yet fully operational, the associated lighting intensity could not be fully captured by the satellite-derived nighttime light data.



Compare with Accessibility

The results show a strong relationship between land cover change and accessibility improvement in the study area. Improvements in accessibility have encouraged rapid urban growth, road expansion, and infrastructure development in nearby districts. This indicates that changes in accessibility are directly influencing the spatial pattern of urbanization in southwestern

Bangladesh. The findings suggest that Padma Bridge is acting as a major driver of regional development and land transformation.



4.2. Limitations of the Classification Framework

While the Random Forest classifier achieved high overall validation accuracy, interpreting the spatial transformation requires acknowledging certain spectral limitations. The three-class schema (Waterbody, Vegetation, Impervious Surface) was occasionally insufficient to capture the nuanced variability of the region's complex surface conditions. Specifically, spectral reflectance overlap led to misclassifications in some areas, most notably confusing sandy riverbanks with impervious urban surfaces, and failing to clearly distinguish between actively cultivated agricultural land and bare soil.

4.3. Future Scope

The current LULC trends suggest that the southwestern region will continue to experience intense urbanization pressure. To better inform regional planning and safeguard food security, future research should employ a more robust multi-class classification system.

Conclusion

This study shows that the opening of the Padma Multipurpose Bridge has significantly influenced land use and economic patterns in southwestern Bangladesh. Between 2017 and 2025, vegetation decreased while impervious surfaces increased, indicating rapid urban expansion. Areas with higher LULC change also show increased nighttime light intensity and improved accessibility, confirming a strong link between infrastructure development, urban growth, and economic activity. Overall, the findings suggest that Padma Bridge is a key driver of regional transformation. However, future planning is necessary to manage rapid land conversion and protect agricultural areas in the region.

Reference

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